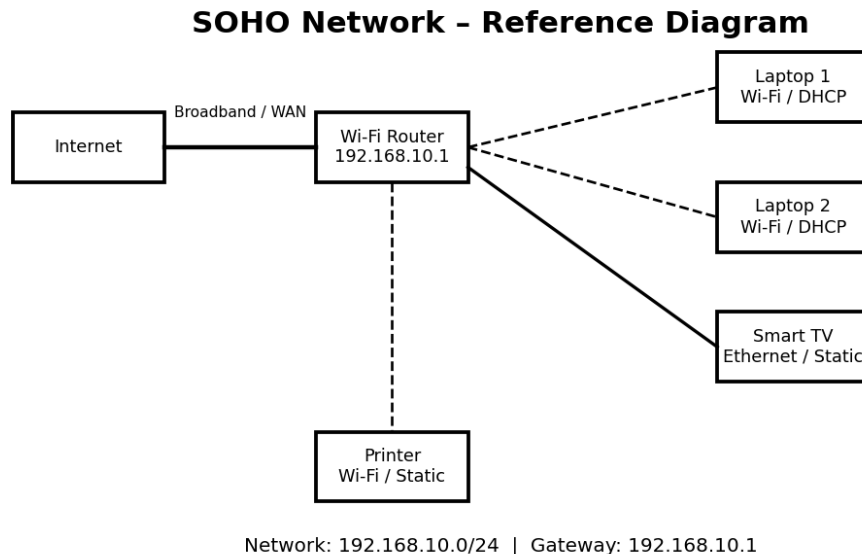


# Build a SOHO network

- 1) Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).

## SOHO Network Diagram



- The reference diagram shows one broadband Internet connection connected to a Wi-Fi router.
- Laptop 1 and Laptop 2 connect wirelessly using Wi-Fi and DHCP.
- The Smart TV uses a wired Ethernet connection and a static IP.
- The Printer uses Wi-Fi and a static IP.

## Addressing Example

- Network: 192.168.10.0/24.
- Router/default gateway: 192.168.10.1.
- Printer: 192.168.10.10 (static).
- Smart TV: 192.168.10.11 (static).
- Laptops: DHCP addresses from the router.

- 2) **List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.**

### **Required Hardware**

- Broadband Internet connection and ISP modem/ONT, depending on the service.
- Wi-Fi router — example: TP-Link Archer AX23/AX55 or ASUS RT-AX53U/RT-AX57 class.
- Two Wi-Fi-capable laptops.
- Smart TV with Wi-Fi or Ethernet.
- Wi-Fi/Ethernet network printer.
- Cat5e or Cat6 Ethernet cables.
- Power adapters and a power strip/UPS.

### **Optional Hardware**

- Unmanaged Gigabit Ethernet switch if additional wired devices are needed.
- Wi-Fi range extender or mesh node if coverage is insufficient.
- Additional Cat6 cables for future devices.

### **Hardware Selection Note**

- Choose the router according to Internet speed, device count, coverage, and available LAN ports.
- Brand/model examples are suggestions; verify current specifications and price before purchasing.

- 3) **Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.**

**Step 1: Connect to the Router**

- Connect a laptop to the router using Wi-Fi or Ethernet.
- Open a browser and enter the router management address shown on the router label/manual.

**Step 2: Change the Admin Password**

- Sign in using the initial administrator credentials.
- Open Administration/System settings.
- Change the default admin password to a strong unique password.
- Save the settings.

**Step 3: Configure Wi-Fi**

- Open Wireless/Wi-Fi settings.
- Set a unique SSID, for example Gattu\_SOHO\_5G.
- Avoid putting personal information in the SSID.

**Step 4: Enable WPA2**

- Select WPA2-Personal/WPA2-PSK where supported.
- Use AES encryption if available.
- Create a strong Wi-Fi password and save.

**Step 5: Configure LAN/DHCP**

- Set LAN address to 192.168.10.1 with subnet 255.255.255.0.
- Enable DHCP for laptops.
- Keep static addresses for the printer and Smart TV outside the DHCP pool or use DHCP reservations.

**Step 6: Verify**

- Reconnect the laptops using the new SSID/password.
- Confirm IP configuration and test Internet access.

- 4) **Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.**

#### **IP Address Plan**

- Router/Gateway: 192.168.10.1.
- Printer: 192.168.10.10 — Static.
- Smart TV: 192.168.10.11 — Static.
- Laptop 1: 192.168.10.100 — DHCP example.
- Laptop 2: 192.168.10.101 — DHCP example.

#### **Device and MAC Address Table**

- Router — MAC: 00:11:22:33:44:01 — IP: 192.168.10.1.
- Printer — MAC: 00:11:22:33:44:10 — IP: 192.168.10.10 (Static).
- Smart TV — MAC: 00:11:22:33:44:11 — IP: 192.168.10.11 (Static).
- Laptop 1 — MAC: 00:11:22:33:44:20 — IP: 192.168.10.100 (DHCP example).
- Laptop 2 — MAC: 00:11:22:33:44:21 — IP: 192.168.10.101 (DHCP example).

#### **Important Note**

- The MAC addresses above are fictional examples as requested. Replace them with actual MAC addresses if you perform the practical.
- DHCP addresses may differ depending on the router DHCP pool.

- 5) Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

**Cause 1: Wi-Fi Connection Problem**

- Check that the laptop is connected to the correct SSID.
- Check signal strength and move closer to the router.
- Disconnect/reconnect Wi-Fi and confirm the adapter is enabled.

**Cause 2: IP/DHCP Problem**

- Run ipconfig.
- Check IP address, subnet mask, default gateway, and DNS.
- Reconnect or renew the DHCP lease.
- Verify the router DHCP service is enabled.

**Cause 3: DNS Problem**

- Ping the gateway: ping 192.168.10.1.
- Test Internet connectivity and use nslookup for a domain.
- If IP connectivity works but names do not resolve, check DNS settings.

**Cause 4: Router/ISP Problem**

- Check whether other devices have Internet access.
- If all devices are offline, check router WAN/Internet status and modem/ONT.
- Restart equipment if appropriate or contact the ISP.

**Troubleshooting Order**

- 1. Wi-Fi → 2. IP/DHCP → 3. Gateway ping → 4. DNS → 5. Internet test → 6. Router/WAN/ISP.